

Commerce XI Maths App

Miscellaneous Exercise 2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 22

Question 1 - Q1

Find the range for the following data:

116, 124, 164, 150, 149, 114, 195, 128, 138, 203, 144

Correct answer: 89

Explanation and steps

Use range = largest value - smallest value. Substitution gives 89.

Question 2 - Q2

Given below is the frequency distribution of weekly wages of 400 workers.

Weekly wages (in ₹00)	10	15	20	25	30	35	40
No. of workers	45	63	102	55	74	36	25

Find the range.

Correct answer: 30

Explanation and steps

Use range = largest value - smallest value. Substitution gives 30.

Question 3 - Q3

Find the range of the following data.

Classes	115-125	125-135	135-145	145-155	155-165	165-175
Frequency	1	4	6	1	3	5

Correct answer: 60

Explanation and steps

Use range = largest value - smallest value. Substitution gives 60.

Question 4 - Q4

The city traffic police issued challans for not observing traffic rules.

Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
No. of challans	40	24	36	58	62	80

Find Q.D.

Correct answer: 16.75

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q3 - Q1)/2$. The verified result is 16.75.

Question 5 - Q5

Calculate Q.D. from the following data.

X (less than)	10	20	30	40	50	60	70
Frequency	5	8	15	20	30	33	35

Correct answer: 12.59

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q3 - Q1)/2$. The verified result is 12.59.

Question 6 - Q6

Calculate the appropriate measure of dispersion for the following open-end data.

Wages (₹)	Less than 35	35-40	40-45	45-50	50-55	55-60
No. of workers	15	50	85	40	27	33

Correct answer: 4.97

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 4.97.

Question 7 - Q7

Calculate Q.D. for the following data.

Height of plants (feet)	2-4	4-6	6-8	8-10	10-12	12-14	14-16
No. of plants	15	20	25	12	18	13	17

Correct answer: 3.25

Explanation and steps

Locate Q1 and Q3, then use $Q.D. = (Q3 - Q1)/2$. The verified result is 3.25.

Question 8 - Q8(i)

For 25, 21, 23, 29, 27, 22, 28, 23, 27, 25,
Find Variance.

Correct answer: 6.6

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 6.6.

Question 9 - Q8(ii)

For 25, 21, 23, 29, 27, 22, 28, 23, 27, 25,
Find Standard Deviation (S. D.)

Options

A. $\sqrt{6.6}$

B. 6.6

C. $\sqrt{5.6}$

D. $\sqrt{7.6}$

Correct answer: $\sqrt{6.6}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{6.6}$.

Question 10 - Q9(i)

Following data gives no. of goals scored by a team in 100 matches.
Compute Variance.

No. of goals scored	0	1	2	3	4	5
No. of matches	15	20	25	15	20	5

Correct answer: 2.16

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 2.16.

Question 11 - Q9(ii)

Following data gives no. of goals scored by a team in 100 matches.
Compute Standard Deviation (S. D.)

No. of goals scored	0	1	2	3	4	5
No. of matches	15	20	25	15	20	5

Options

A. $\sqrt{2.16}$

B. 2.16

C. $\sqrt{1.16}$

D. $\sqrt{3.16}$

Correct answer: $\sqrt{2.16}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{2.16}$.

Question 12 - Q10(i)

C.I.	45-55	55-65	65-75	75-85	85-95	95-105
f	4	2	5	3	6	5

Compute arithmetic mean.

(Given $\sqrt{(296)} = 17.20$)

Correct answer: 78

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 78.

Question 13 - Q10(ii)

C.I.	45-55	55-65	65-75	75-85	85-95	95-105
f	4	2	5	3	6	5

Compute S.D. correct to two decimal places.

(Given $\sqrt{(296)} = 17.20$)

Correct answer: 17.20

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 17.20.

Question 14 - Q10(iii)

C.I.	45-55	55-65	65-75	75-85	85-95	95-105
f	4	2	5	3	6	5

Compute C.V. as a percentage correct to two decimal places.

(Given $\sqrt{(296)} = 17.20$)

Correct answer: 22.05%

Explanation and steps

Use $C.V. = (S.D./\text{mean}) \times 100$ and compare the percentages when required. The verified result is 22.05%.

Question 15 - Q11(i)

The mean and S.D. of 200 items were calculated as 60 and 20 respectively.

Two items were wrongly taken as 3 and 67 instead of 13 and 17.

Find the correct mean.

Correct answer: 59.8

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 59.8.

Question 16 - Q11(ii)

The mean and S.D. of 200 items were calculated as 60 and 20 respectively.
Two items were wrongly taken as 3 and 67 instead of 13 and 17.
Find the correct variance.

Correct answer: 403.76

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 403.76.

Question 17 - Q12(i)

The mean and S.D. of 48 observations are 40 and 8 respectively.
If two more observations 60 and 65 are added to set.
Find the mean of 50 observations.

Correct answer: 40.9

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is 40.9.

Question 18 - Q12(ii)

The mean and S.D. of 48 observations are 40 and 8 respectively.
If two more observations 60 and 65 are added to set.
Find the Standard Deviation of 50 items.

Options

- A. $\sqrt{81.13}$
- B. 81.13
- C. $\sqrt{80.13}$
- D. $\sqrt{82.13}$

Correct answer: $\sqrt{81.13}$

Explanation and steps

Apply the appropriate variance formula and take its positive square root when S.D. is requested. The verified result is $\sqrt{81.13}$.

Question 19 - Q13

The mean height of 200 students is 65 inches.

The mean heights of boys and girls are 70 inches and 62 inches respectively and the standard deviations are 8 and 10 respectively.

Find the number of boys and the combined S.D.

Options

A. 75 Boys, Combined S.D. $\sqrt{(101.5)}$

B. 125 Boys, Combined S.D. $\sqrt{(101.5)}$

C. 75 Boys, Combined S.D. 101.5

D. 125 Boys, Combined S.D. $\sqrt{(100.5)}$

Correct answer: 75 Boys, Combined S.D. $\sqrt{(101.5)}$

Explanation and steps

First find the size-weighted combined mean, then include both within-group variance and the squared difference of each group mean from the combined mean. This gives 75 Boys, Combined S.D. $\sqrt{(101.5)}$.

Question 20 - Q14

From the following data available for 5 pairs of observations of two variables x and y, obtain the combined S.D. for all 10 observations.

Where, $\Sigma x_i = 30$, $\Sigma y_i = 40$, $\Sigma x_i^2 = 225$, $\Sigma y_i^2 = 340$

Options

A. $\sqrt{(7.5)}$

B. 7.5

C. $\sqrt{(6.5)}$

D. $\sqrt{(8.5)}$

Correct answer: $\sqrt{(7.5)}$

Explanation and steps

First find the size-weighted combined mean, then include both within-group variance and the squared difference of each group mean from the combined mean. This gives $\sqrt{(7.5)}$.

Question 21 - Q15

The mean and standard deviations of two brands of watches are given below:

	Brand-I	Brand-II
Mean	36 months	48 months
S.D.	8 months	10 months

Calculate coefficient of variation for the two brands and interpret the results.

Options

- A. $CV_{(B-I)} = 22.22\%$, $CV_{(B-II)} = 20.83\%$, Brand-I is more variable
- B. $CV_{(B-I)} = 20.83\%$, $CV_{(B-II)} = 22.22\%$, Brand-II is more variable
- C. $CV_{(B-I)} = 22.22\%$, $CV_{(B-II)} = 20.83\%$, Brand-II is more variable
- D. $CV_{(B-I)} = 20.83\%$, $CV_{(B-II)} = 22.22\%$, Brand-I is more variable

Correct answer: $CV_{(B-I)} = 22.22\%$, $CV_{(B-II)} = 20.83\%$, Brand-I is more variable

Explanation and steps

$CV(\text{Brand-I}) = (8/36) \times 100 = 22.22\%$ and $CV(\text{Brand-II}) = (10/48) \times 100 = 20.83\%$. The larger coefficient of variation indicates greater variability, so Brand-I is more variable and Brand-II is more consistent. Textbook answer note: the key's statement that Brand-II is more variable conflicts with these calculated C.V. values and with the definition of C.V.; the app uses the mathematically correct interpretation.

Question 22 - Q16

C.I.	5-15	15-25	25-35	35-45	45-55	55-65	65-75
f	6	7	15	25	8	18	21

Calculate coefficient of variation as a percentage correct to two decimal places.

Correct answer: 39.49%

Explanation and steps

Use $C.V. = (S.D./\text{mean}) \times 100$ and compare the percentages when required. The verified result is 39.49%.